

package Program1;

import java.util.Scanner;

public class complexnuminp {
 double real, img;

complexnuminp(double r, double i) {
 real = r;
 img = i;
 }
}

public static complexnuminp sum(complexnuminp c1, complexnuminp c2) {
 complexnuminp result = new complexnuminp(0, 0);
 result.real = c1.real + c2.real;
 result.img = c1.img + c2.img;
 return result;
 }
}

public static void main(String args[]) {
 double r1, i1, r2, i2;
 Scanner sc = new Scanner(System.in);

 System.out.print("Enter The Real Part of c1 : ");
 r1 = sc.nextDouble();
 System.out.print("Enter The Imaginary Part of c1 : ");
 i1 = sc.nextDouble();

System.out.print("Enter The Real Part of c2 : ");
 r2 = sc.nextDouble();
 System.out.print("Enter The Imaginary Part of c2 : ");
 i2 = sc.nextDouble();

complexnuminp c1 = new complexnuminp(r1, i1);
 complexnuminp c2 = new complexnuminp(r2, i2);

complexnuminp sumResult = sum(c1, c2);

System.out.println("Sum is : " + sumResult.real + " + " + sumResult.img + "i");

sc.close();
 }
}

package Program2;

```
public class product {

    String pname, pcode;
    int price;

    public product() {}

    public product(String pname, String pcode, int price) {
        this.pname = pname;
        this.pcode = pcode;
        this.price = price;
    }

    public String getPname() {    return pname;  }
    public void setPname(String pname) {
        this.pname = pname;
    }

    public String getPcode() {    return pcode;  }
    public void setPcode(String pcode) {    this.pcode = pcode;  }

    public int getPrice() {    return price;    }
    public void setPrice(int price) {    this.price = price;  }

    public void display() {
        System.out.println("pcode : " + this.pcode);
        System.out.println("pname : " + this.pname);
        System.out.println("price : " + this.price + "\n");
    }

    public static void main(String[] args) {
        product p1 = new product();
        p1.pcode = "car123";
        p1.pname = "benz";
        p1.price = 10000;
        System.out.println("displaying p1:");
        p1.display();

        product p2 = new product("jaguar", "car426", 25000);
        System.out.println("displaying p2:");
        p2.display();

        product p3 = new product("maruthi", "car800", 50000);
        System.out.println("displaying p3:");
        p3.display();

        product lowest = p3.price < (p1.price < p2.price ? p1.price : p2.price) ? p3 : (p1.price < p2.price ? p1 : p2);

        System.out.println("displaying product with lowest price:");
        lowest.display();
    }
}
```

package Program3;**import java.util.Scanner;**

```
public class matrixaddition {
    public static void main(String[] args) {
        int r1, c1, r2, c2;
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number of Rows in Matrix A: ");
        System.out.print("Enter Number of Columns in Matrix A: ");
        System.out.print("Enter Number of Rows in Matrix B: ");
        System.out.print("Enter Number of Columns in Matrix B: ");

        r1 = sc.nextInt();
        c1 = sc.nextInt();
        r2 = sc.nextInt();
        c2 = sc.nextInt();

        if (r1 == r2 && c1 == c2) {
            int matrixA[][] = new int[r1][c1];
            int matrixB[][] = new int[r2][c2];

            System.out.println("Enter The Elements of Matrix A:");
            for (int i = 0; i < r1; i++) {
                for (int j = 0; j < c1; j++) {
                    matrixA[i][j] = sc.nextInt();
                }
            }

            System.out.println("Enter The Elements of Matrix B:");
            for (int i = 0; i < r2; i++) {
                for (int j = 0; j < c2; j++) {
                    matrixB[i][j] = sc.nextInt();
                }
            }

            System.out.println("\nMatrix A:");
            for (int i = 0; i < r1; i++) {
                for (int j = 0; j < c1; j++) {
                    System.out.print(matrixA[i][j] + " ");
                }
                System.out.println();
            }

            System.out.println("Matrix B:");
            for (int i = 0; i < r2; i++) {
                for (int j = 0; j < c2; j++) {
                    System.out.print(matrixB[i][j] + " ");
                }
                System.out.println();
            }

            System.out.println("The Sum Matrix:");
            for (int i = 0; i < r1; i++) {
                for (int j = 0; j < c1; j++) {
                    System.out.print((matrixA[i][j] + matrixB[i][j]) + " ");
                }
                System.out.println();
            }
        } else {
            System.out.println("These Matrices Cannot be Added (Dimensions must match).");
        }

        sc.close();
    }
}
```

package Program4;

```
import java.util.Scanner;
```

```
public class MatrixSymmetry {
```

```
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
  
        System.out.println("Enter the No of rows:");  
        int rows = sc.nextInt();  
        System.out.println("Enter the No of columns:");  
        int cols = sc.nextInt();
```

```
  
        int matrix[][] = new int[rows][cols];
```

```
  
        System.out.println("Enter the Elements of the matrix:");  
        for (int i = 0; i < rows; i++) {  
            for (int j = 0; j < cols; j++) {  
                matrix[i][j] = sc.nextInt();  
            }  
        }  
    }
```

```
  
        System.out.println("The Given Matrix:");  
        for (int i = 0; i < rows; i++) {  
            for (int j = 0; j < cols; j++) {  
                System.out.print(matrix[i][j] + "\t");  
            }  
            System.out.println();  
        }
```

```
  
        if (rows != cols) {  
            System.out.println("The Given Matrix is not a Square");  
        } else {  
            boolean symmetric = true;  
            for (int i = 0; i < rows; i++) {  
                for (int j = 0; j < cols; j++) {  
                    if (matrix[i][j] != matrix[j][i]) {  
                        symmetric = false;  
                        break;  
                    }  
                }  
            }  
            if (!symmetric) {  
                break;  
            }  
        }
```

```
  
        if (symmetric) {  
            System.out.println("It is symmetric");  
        } else {  
            System.out.println("It is not symmetric");  
        }  
    }
```

```
    sc.close();
```

```
    }  
}
```

package Program5;

```
import java.util.Scanner;
```

```
public class CPU {  
    double price;
```

```
    class Processor {  
        double cache;  
        String manufacturer;
```

```
        Processor(double cache) {  
            this.cache = cache;  
        }
```

```
        double getCache() {  
            return cache;  
        }
```

```
    }
```

```
    static class RAM {  
        double clockSpeed;  
        String manufacturer;
```

```
        RAM(double speed) {  
            this.clockSpeed = speed;  
        }
```

```
        double getClockSpeed() {  
            return clockSpeed;  
        }
```

```
    }
```

```
    public static void main(String[] args) {  
        CPU myCpu = new CPU();  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Enter cache of your processor: ");  
        double cacheVal = sc.nextDouble();
```

```
        System.out.println("Enter RAM ClockSpeed: ");  
        double speedVal = sc.nextDouble();
```

```
        CPU.Processor proc = myCpu.new Processor(cacheVal);  
        CPU.RAM ram = new CPU.RAM(speedVal);
```

```
        System.out.println("Processor Cache: " + proc.getCache() + " MB");  
        System.out.println("RAM Clock Speed: " + ram.getClockSpeed() + " GHz");
```

```
        sc.close();
```

```
    }
```

```
}
```

package Program7;

```
import java.util.Scanner;
```

```
public class LinearSearch {  
    public static void main(String[] args) {  
        int i, count, target, arr[];  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Enter Number of Elements: ");  
        count = sc.nextInt();  
        arr = new int[count];  
  
        System.out.println("Enter Those " + count + " elements:");  
        for (i = 0; i < count; i++) {  
            arr[i] = sc.nextInt();  
        }  
  
        System.out.println("Enter Value You Want to Find: ");  
        target = sc.nextInt();  
  
        for (i = 0; i < count; i++) {  
            if (arr[i] == target) {  
                System.out.println(target + " is present at location: " + (i + 1));  
                break;  
            }  
        }  
  
        if (i == count) {  
            System.out.println(target + " is not present in the array.");  
        }  
  
        sc.close();  
    }  
}
```

package Program10;

import java.util.Scanner;

public class OverLoad {

void area(float side) {

 System.out.println("The area of square: " + Math.pow(side, 2) + " sq units");

}

void area(float length, float breadth) {

 float result = length * breadth;

 System.out.println("The area of the rectangle is: " + result + " sq units");

}

void area(double radius) {

 double result = 3.14 * radius * radius;

 System.out.println("The area of the circle is: " + result + " sq units");

}

public static void main(String args[]) {

 OverLoad obj = new OverLoad();

 Scanner sc = new Scanner(System.in);

 System.out.print("Enter The Side of Square: ");

 float s = sc.nextFloat();

 obj.area(s);

 System.out.print("Enter The Length of Rectangle: ");

 float l = sc.nextFloat();

 System.out.print("Enter The Breadth of Rectangle: ");

 float b = sc.nextFloat();

 obj.area(l, b);

 System.out.print("Enter The Radius of Circle: ");

 double r = sc.nextDouble();

 obj.area(r);

 sc.close();

}

}

package Program11;

import java.util.Scanner;

class EmployeeT {

int empid;
String name;
float salary;
String address;

EmployeeT() {}

EmployeeT(int empid, String name, float salary, String address) {

this.empid = empid;
this.name = name;
this.salary = salary;
this.address = address;

}

}

class Teacher extends EmployeeT {

String department, subject;

Teacher(int empid, String name, float salary, String address, String department, String subject) {

super(empid, name, salary, address);
this.department = department;
this.subject = subject;

}

public void display() {

System.out.println("Teacher ID: " + empid);
System.out.println("Teacher Name: " + name);
System.out.println("Teacher Salary: " + salary);
System.out.println("Teacher Address: " + address);
System.out.println("Teacher Department: " + department);
System.out.println("Teacher Subject: " + subject);

}

}

public class TeacherArrObjects {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.println("Enter Number of Teachers:");

int count = sc.nextInt();

Teacher list[] = new Teacher[count];

for (int i = 0; i < count; i++) {

System.out.println("\nEnter Details for Teacher " + (i + 1) + " :");

System.out.print("Enter ID (integer): ");

int tid = sc.nextInt();

System.out.print("Enter Name (String): ");

String name = sc.next();

System.out.print("Enter Salary (float): ");

float salary = sc.nextFloat();

System.out.print("Enter Address (String): ");


```
String address = sc.next();

System.out.print("Enter Department (String): ");
String department = sc.next();

System.out.print("Enter Subject (String): ");
String subject = sc.next();

// Creating the object and storing it in the array
list[i] = new Teacher(tid, name, salary, address, department, subject);
}

System.out.println("\n--- Teachers List ---");
for (Teacher t : list) {
    t.display();
    System.out.println();
}

sc.close();
}
}
```

```
package Program12;
```

```
import java.util.Scanner;
```

```
interface AP{  
    void input();  
    void area();  
    void perimeter();  
}
```

```
class Circle implements AP{  
    int r = 0;  
    double pi = 3.14, area = 0, perimeter = 0;  
    public void input() {  
        Scanner c = new Scanner(System.in);  
        r = c.nextInt();  
    }  
  
    public void area() {  
        area = pi * r * r;  
        System.out.println("Area of circle : " +area);  
    }  
  
    public void perimeter() {  
        perimeter = 2 * pi * r;  
        System.out.println("Perimeter of circle : "+perimeter);  
    }  
}
```

```
class Rectangle implements AP{  
    int l=0, b=0;  
    double area,perimeter;  
    public void input() {  
        Scanner r = new Scanner(System.in);  
        l = r.nextInt();  
        b = r.nextInt();  
    }  
    public void area() {  
        area = l * b;  
        System.out.println("Area of rectangle : " + area);  
    }  
    public void perimeter() {  
        perimeter = 2 * (l+b);  
        System.out.println("Perimeter of rectangle : " +perimeter);  
    }  
}
```

```
public class IFaceDemo {  
    public static void main(String[] args) {  
        int ch;  
        Circle c = new Circle();  
        Rectangle r = new Rectangle();  
        Scanner sc = new Scanner(System.in);  
  
        System.out.println("Interface Menu - Area & Perimeter of Circle & Rectangle ");  
        System.out.println("1: Area of Circle\n2: Area of Rectangle\n3: Perimeter of Circle \n4: Perimeter of Rectangle\n5:  
Program Termination");  
  
        lp : while(true) {  
            System.out.println("Make your choice:");
```

```

ch = sc.nextInt();
switch (ch) {
case 1:
    System.out.println("Enter the radius of the circle \n");
    c.input();
    c.area();
    break;

case 2:
    System.out.println("Enter the length and breadth of the rectangle");
    r.input();
    r.area();
    break;

case 3:
    System.out.println("Enter the radius of the circle \n");
    c.input();
    c.perimeter();
    break;

case 4:
    System.out.println("Enter the length and breadth of the rectangle \n");
    r.input();
    r.perimeter();
    break;

case 5:
    System.out.println("Exiting ... \n");
    break lp;

default:
    System.out.println("Invalid choice !!! \nPlease make a valid choice...");

}
}

sc.close();
}
}

```

package Program13;

import java.util.Scanner;

class Person {

String name, gender, address;

int age;

Person(String name, String gender, String address, int age) {

this.name = name;

this.gender = gender;

this.address = address;

this.age = age;

}

}

class Employee extends Person {

String empid, company_name, qualification;

float salary;

Employee(String name, String gender, String address, int age, String empid, String company, String qual, float salary) {

super(name, gender, address, age);

this.empid = empid;

this.company_name = company;

this.qualification = qual;

this.salary = salary;

}

}

class Teacher extends Employee {

String subject, department, teacherid;

Teacher(String name, String gender, String address, int age, String empid, String company, String qual, float salary, String subject, String department, String tid) {

super(name, gender, address, age, empid, company, qual, salary);

this.subject = subject;

this.department = department;

this.teacherid = tid;

}

public void display() {

System.out.println("Teacher ID: " + teacherid);

System.out.println("Employee ID: " + empid);

System.out.println("Name: " + name);

System.out.println("Gender: " + gender);

System.out.println("Address: " + address);

System.out.println("Age: " + age);

System.out.println("Company: " + company_name);

System.out.println("Qualification: " + qualification);

System.out.println("Salary: " + salary);

System.out.println("Subject: " + subject);

System.out.println("Department: " + department);

}

}

public class TeacherArrObjects {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.println("Enter the number of teachers:");

int count = sc.nextInt();

```

Teacher list[] = new Teacher[count];

for (int i = 0; i < count; i++) {
    System.out.println("\n--- Enter Details for Teacher " + (i + 1) + " ---");

    System.out.print("Teacher ID: ");
    String tid = sc.next();

    System.out.print("Employee ID: ");
    String empid = sc.next();

    System.out.print("Name: ");
    String name = sc.next();

    System.out.print("Gender: ");
    String gender = sc.next();

    System.out.print("Address: ");
    String address = sc.next();

    System.out.print("Age: ");
    int age = sc.nextInt();

    System.out.print("Company Name: ");
    String company = sc.next();

    System.out.print("Department: ");
    String dept = sc.next();

    System.out.print("Qualification: ");
    String qual = sc.next();

    System.out.print("Subject: ");
    String sub = sc.next();

    System.out.print("Salary: ");
    float sal = sc.nextFloat();

    list[i] = new Teacher(name, gender, address, age, empid, company, qual, sal, sub, dept, tid);
}

System.out.println("\n--- Displaying Teachers ---");
for (Teacher t : list) {
    t.display();
    System.out.println("-----");
}

sc.close();
}
}

```